Evaluating Terminology Translation in Machine Translation Systems via Metamorphic Testing

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ABSTRACT

Machine translation has become an integral part of daily life, with terminology translation playing a crucial role in ensuring the accuracy of translation results. However, existing translation systems, such as Google Translate, have been shown to occasionally produce errors in terminology translation. Current metrics for assessing terminology translation rely on reference translations and bilingual dictionaries, limiting their effectiveness in large-scale automated MT system testing.

To address this challenge, we propose a novel method: Metamorphic Testing for Terminology Translation (TermMT), which achieves effective and efficient testing for terminology translation in MT systems without relying on reference translations or bilingual terminology dictionaries. Our approach involves constructing metamorphic relations based on the characteristics of terms: (a) adding an appropriate reference of the term in the given context would *not change* the translation of the term; (b) if we modify part of a multi-word term, the translation of the revised word combination would *change*. To evaluate the effectiveness of TermMT, we tested the terminology translation capabilities of three machine translation systems, Google Translate, Bing Microsoft Translator, and mBART, using the English portion of the bilingual UM-corpus dataset. The results show that TermMT detected a total of 3,765 translation errors on Google Translate, 2,351 on Bing Microsoft Translator, and 6,011 on mBART, with precisions of 82.33%, 83.00%, and 86.33%, respectively.

KEYWORDS

Software Testing, Machine Translation, Terminology Translation, Metamorphic Testing

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1 INTRODUCTION

Due to the scarcity of professional translators and the increasing need for immediate translation [28], Machine Translation (MT) systems have experienced rapid advancements over the last decades, gaining widespread popularity. Mainstream MT systems, such as Google Translate [18], processed over 100 billion words daily in 2018 [44] and had amassed over 1 billion users by 2023 [19]. Although these mainstream MT systems, primarily built on neural MT [35], have shown their strengths, they inevitably contain flaws and can lead to severe consequences. To illustrate, Google Translate mistranslated a sign from Welsh, rendering "Blasting in Progress" as "workers exploding" [13]. Moreover, MT systems are prone to contain errors more frequently than human translators [37], underscoring the urgent need for rigorous testing of MT systems.

Recently, researchers have designed and implemented a series of test works targeting MT systems [37, 48]. The central premise of these efforts revolves around mitigating the lack of test oracles [7] by deploying input mutations and metamorphic rules [36], subsequently assessing the consistency [12, 24] and robustness [22] of the MT systems. As research endeavors delve deeper, certain fine-grained features of machine translation systems have come into the purview of researchers as subjects for testing. Wang et al. [42] proposed a testing framework predicated on four novel mutation operators for assessing the word sense disambiguation capabilities of MT systems. Their work demonstrates that devising testing methodologies tailored to the fine-grained functionalities of MT systems can unearth unique errors.

Inspired by Wang et al.'s work [42], this paper aims to examine another vital fine-grained aspect of MT: terminology translation. Terms, which serve as labels or designations for fundamental concepts and their relationships within a specific field or subfield [10], are crucial in language translation, such as between English and Chinese. The precision of terminology translation is paramount in MT systems and often directly influences the overall translation quality. However, errors in terminology translation are common in



Figure 1: Translation error of Google Translate collected on May 24, 2024, which translates the term "part of speech" into Chinese with the meaning "part of pronunciation".

machine translation systems. Figure 1 illustrates actual translation errors of commonly used terms extracted from Google Translate, where the term "part of speech" in the original English sentence is incorrectly translated into Chinese as "part of pronunciation." This erroneous translation completely alters the meaning of the original sentence. This example highlights the critical need for effective methods to detect and address terminology translation errors in MT systems.

In terminology translation, translators' proficiency is often assessed using specific translation metrics (e.g., exact-match accuracy [4], lemmatised term exact match accuracy [9], and TermEval [21]). These metrics rely heavily on reference translations and reliable bilingual terminology dictionaries [20], which *limits their use in large-scale automated testing of MT systems*. To tackle the above problem, this paper proposes a novel testing method, Metamorphic Testing for Terminology Translation (**TermMT** for short), to achieve effective and efficient testing for terminology translation in MT systems. The main idea of our method lies in designing appropriate metamorphic relations (MRs), i.e., formulating corresponding MRs based on the characteristics of terms. Specifically, we focus on two aspects. (a) We construct MRs on the following **unchangeable** restriction: adding an appropriate reference [45] (e.g., "a linguistic category of words sharing syntactic or morphological behavior and semantic properties" is a reference for the term "Part of Speech") of the term in the given context would *not change* the translation of the term. (b) Unlike previous studies [12, 22, 24, 36, 37, 42, 48], we employ the following **changeable** restriction to design MRs. Since multi-word terms usually are with a stable word combination [8], the translation of the revised word combination (e.g., replacing one word with another word) would *change* if we modify part of a multi-word term, e.g., the terms "Part of Speech" and its mutant "System of Speech" (by converting "Part" into "System") are obvious with different meanings and expected to have different translations.

Our method TermMT includes the following steps (see Section 3 and Figure 3 for details). (a) Pre-processing: We construct a term dictionary for annotation and mutation from existing terminology dictionaries IATE [23] and Wiktionary [43] and select original sentences that contain single multi-word terms for mutation. (b) Mutation: To achieve metamorphic relationships, we design two mutation operators, meaning insertion and term partial replacement, which generate mutated sentences corresponding to the original sentence for MR violation detection. (c) Translation and Alignment: We input the original and mutated sentences into the translation system to obtain the translation results, then align the source sentences with the translations using the widely used alignment model AWESOME [17] to extract the translations of the terms in both the original and mutated sentences. (d) Error Detection: We design three novel metamorphic relations (MRs) to describe the unchangeable and changeable restriction motioned above and use BGE [46] to calculate the semantic similarity of the terms before and after translation and the translation results. We identify term translation errors by detecting MR violations based on semantic similarity.

To evaluate the effectiveness of our method in detecting translation errors, we use the English portion of five professional fields from the bilingual dataset UM-Corpus [40], which includes data from multiple domains, as the sentence source. Through terminology annotation, we select more than 110 thousand sentences as the original set for mutation. We conduct our experiments on the English-Chinese translation setting using two commercial MT systems, Google Translate [18] and Bing Microsoft Translator [30], as well as an open-source translator, mBART [38]. The results show that TermMT detects a total of 3,765 term translation errors on Google Translate, 2,351 on Bing Microsoft Translator, and 6,011 on mBART, with precisions of 82.33%, 83.00%, and 86.33%, respectively. Additionally, we compare TermMT with the state-of-the-art translation error detection method, CAT [37]. The result shows that over 80% of the errors identified by TermMT could not be detected by CAT.

The main contributions of this paper are as follows.

- **Scenario.** We pioneer large-scale testing for terminology translation abilities of MT systems, which could be a core issue in translation, particularly in professional contexts.
- **Strategy.** We introduce a novel testing method, Metamorphic Testing for Terminology (TermMT), to identify translation errors by employing three innovative metamorphic relations to detect terminology translation errors.
- **Study.** We conduct an extensive empirical study from five professional fields on three translation systems. The results demonstrate that our method can detect a significant number of translation errors with an accuracy exceeding 80% across all three systems.

2 BACKGROUND AND MOTIVATION

2.1 Terminology in Translation

Terminology, as a subject, encompasses the theory regarding the distinction between terms and words, and as a type of vocabulary, it refers to the lexicon of a specific subject field [11]. According to the General Terminology Theory [45], a term can be distinguished from words in ordinary language by its single-meaning relationship with the specialized concept it denotes. According to Bowker and Pearson [11], terms can be single words but are often composed of multiple words. Due to its nature, terminology is considered one of the most critical areas of specialized translation [39].

Several term databases have been developed to manage terminology better and facilitate translation work, such as IATE (Inter-Agency Terminology Exchange) [25], the central term database for all bodies of the European Union. These databases serve as repositories of specialized vocabulary, aiding translators in maintaining consistency and accuracy across their translations.

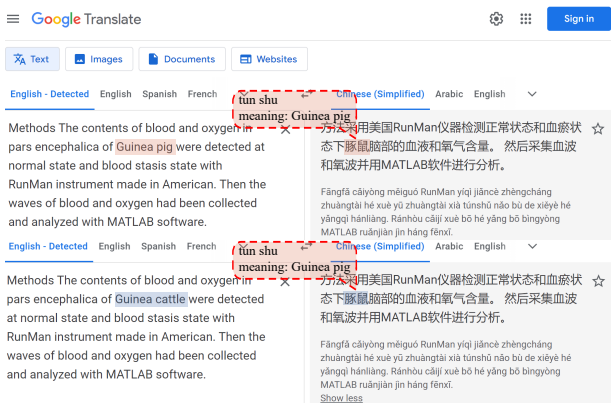


Figure 2: An Example of term translation errors on Google Translate collected on May 25, 2024

Researchers have proposed various methods in neural machine translation to enhance the model's ability to translate terminology, such as terminology constraints [16] and terminology integration [29]. A model's terminology translation capabilities are typically assessed through custom machine metrics combined with reference translation results, like exact-match accuracy [4].

2.2 Motivation Example

This section will showcase an actual translation error found on Google Translate and explain how our method detects it. As illustrated in Figure 2, an error is observed in the translation provided by Google Translate. Initially, we input the original sentence into the translation system. In the original sentence, the multi-word term "Guinea pig" is correctly translated by Google Translate as "tun shu" in Chinese (which means "Guinea pig"). Afterward, we modify the term by replacing the word "pig" with "cattle" and input this mutated sentence into the translation system. Surprisingly, Google Translate still translates it as "tun shu" despite the change, which demonstrates a terminology translation error. Here we employ this case to illustrate the advantage of our method.

Reference-based methods [4, 9, 21] cannot detect this error because they only use the original sentence, and the translation of the original sentence is correct. Moreover, existing MT testing methods, such as CAT [37], could not identify this error, as they only consider unchangeable restrictions. These methods would flag no error because the translation results of both sentences are precisely the same.

On the contrary, with our method, we posit the MR that replacing one word in a multi-word term should result in a mutated phrase with *the different meaning and translation* from the original term. In this example, our method successfully detects an MR violation and identifies the error, as both translations of the original and mutated sentences are the same. This case demonstrates our method's practical application in detecting MT errors.

3 APPROACH

This section introduces our method of four parts, illustrated in Figure 3. We will subsequently explain each part comprehensively.

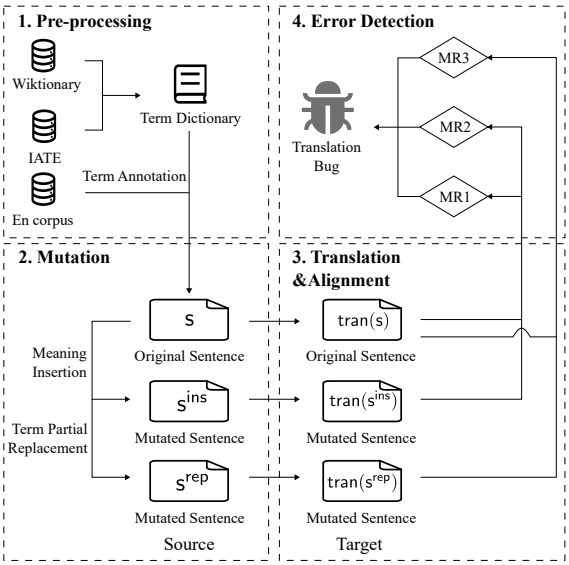


Figure 3: The Overall Framework of TermMT

3.1 Pre-processing

We first construct a term dictionary and then collect original sentences for mutation based on this dictionary.

(a) We use overlapping parts of English multi-word terms from the terminology datasets Wiktionary [43] and IATE [23] as terminology sources. Beyond the terms themselves, we extract domain information from IATE and the meaning, part of speech, alternative forms, synonyms, and Chinese translations from Wiktionary to construct a comprehensive term dictionary. To ensure grammatical correctness in subsequent mutants and the translatability of terms, we select only terms that include noun meanings and Chinese translations. Due to the relatively free grammar in Wiktionary term meanings, we have performed manual grammar checks and corrections on the dictionary entries to ensure quality. Ultimately, we have obtained a term dictionary containing 4,518 multi-word terms.

(b) Leveraging the constructed term dictionary, we annotate the English corpus to identify original sentences containing a single multi-word noun term. We exclude sentences exceeding 50 tokens due to their excessive length[1].

3.2 Mutation

We design two mutation operators for original sentences to generate mutated sentences: meaning insertion and term partial replacement. Below, we provide detailed explanations of these two mutation operators.

3.2.1 Meaning Insertion. Given an original sentence containing a single multi-word term, the goal of meaning insertion is to find the term's meaning in the context of the current sentence from the dictionary and insert it into the sentence.

For an original sentence s containing a multi-word term t, we first use a part-of-speech analysis tool flair [3] to analyze t, which

[1]Section 4.2 and Table 2 summarize the final extracted results.

Table 1: An Example Illustrating TermMT's Mutation, Translation, and Alignment Result

	English Sentence (Source)	Chinese Sentence (Target)	Meaning of Target Sentence
Original Sentence	A Study of **Part of Speech** Classification of Mongolian Language for Information Processing	蒙古语信息处理**部分语音**分类研究	A Study on the Classification of **Part of Pronunciation** in Mongolian Information Processing
Mutated Sentence (Meaning Insertion)	A Study of **Part of Speech** (a linguistic category of words sharing syntactic or morphological behaviour and semantic properties, such as noun or verb) Classification of Mongolian Language for Information Processing	**词性**研究（共享句法或形态行为和语义属性的单词的语言类别，例如名词或动词）用于信息处理的蒙古语分类	A **Part of Speech** study (linguistic categories of words that share syntactic or morphological behavior and semantic properties, such as nouns or verbs) is used to classify Mongolian words for information processing
Mutated Sentence (Term Partial Replacement)	A Study of **System of Speech** Classification of Mongolian Language for Information Processing	面向信息处理的蒙古语**语音**分类**系统**研究	A Study on Mongolian **Pronunciation** Classification **System** for Information Processing

enhances the precision of dictionary matching. Then, we search the dictionary for the meaning list: $\mathsf{mean}_1, ..., \mathsf{mean}_n$ of t under this part-of-speech. To make the interpretation most consistent with the contextual meaning, we add the meaning to the original sentence as bracket annotations and obtain the candidate mutant set $\mathsf{Candis}^{\mathsf{ins}}$. Next, we employ the language model BGE [46], which achieves state-of-the-art performance on both the MTEB [31] (English embedding benchmark) and the C-MTEB [46] (Chinese embedding benchmark) leaderboards, to characterize the original sentence and its mutants, identifying the mutated sentence with the highest cosine similarity to the original sentence as the final mutation result. The calculation formula is as follows:

$$\mathsf{s}^{\mathsf{ins}} = \underset{\mathsf{candi}_i^{\mathsf{ins}} \in \mathsf{Candis}^{\mathsf{ins}}}{\arg\max} \ \mathsf{cos_simi}(\mathsf{s}, \mathsf{candi}_i^{\mathsf{ins}})$$

where $\mathsf{cos_simi}()$ is the function to calculate the cosine similarity, and $\mathsf{s}^{\mathsf{ins}}$ is the mutated sentence with the highest cosine similarity to the original sentence s.

An example of meaning insertion is provided in Table 1 and Figure 3. For the original sentence containing the multi-word term "Part of Speech", we extract the contextual meaning of the term from the dictionary. This meaning is then inserted into the original sentence as a bracketed explanation, resulting in a meaning insertion mutated sentence.

3.2.2 Term Partial Replacement. In the term partial replacement mutation stage, we replace a component word of the multi-word term in the original sentence to generate phrases with different meanings. For the original sentence s containing the term t, we use flair to perform part-of-speech analysis and select all noun positions as the candidate replacement target set. We then refer to [37] and use the BERT model to select appropriate candidate words. Specifically, for each target position, we use a "[MASK]" token to create a masked sentence set, input these into BERT to obtain sentence vectors, and then use a pre-trained linear classifier to get an initial set of candidate words $\mathsf{W}_i^{\mathsf{rep}}$.

We adhere to the settings outlined in [37] to filter candidate words. Utilizing a lemmatization tool, we exclude words sharing the same root as the original and retain only nouns for grammatical correctness. Additionally, by querying the dictionary, we filter out words that could form alternative forms or synonyms of the original term, resulting in the candidate word set $\mathsf{W}_i^{\mathsf{rep}}$ and the corresponding mutated sentence set $\mathsf{S}_i^{\mathsf{rep}}$.

The final term partial replacement mutated sentence set $\mathsf{S}^{\mathsf{rep}}$ of the original sentence s is obtained by merging all $\mathsf{S}_i^{\mathsf{rep}}$: $\mathsf{S}^{\mathsf{rep}} = \bigcup_{i=1}^{n} \mathsf{S}_i^{\mathsf{rep}}$. Each sentence in the final set of mutated sentences $\mathsf{s}^{\mathsf{rep}}$ and the phrases $\mathsf{t}^{\mathsf{rep}}$ are derived after the partial replacement of the terms they contain.

In Table 1 and Figure 3, we provide an example of the term partial replacement. For the original sentence containing the multi-word term "Part of Speech", we replace the word "Part" within the term with "System", resulting in a term partial replacement mutated sentence. This sentence contains the phrase "System of Speech," which has a different meaning from the original term.

3.3 Translation and Alignment

After completing the two types of mutations, we translate the original sentence and the two kinds of mutated sentences. Next, we align the source sentences and their translations using the pre-trained alignment model to extract the translation results of the terms within the sentences.

For an original sentence s containing a multi-word term t, the mutated sentence is s^{x}, where $\mathsf{x} \in \mathsf{ins}, \mathsf{rep}$. After processing through the translation system, the resulting translations are $\mathsf{tran}(\mathsf{s})$ and $\mathsf{tran}(\mathsf{s}^{\mathsf{x}})$. We then align these translations and obtain the translation results of the terms and mutated phrases: $\mathsf{tran}(\mathsf{t}, \mathsf{s})$, $\mathsf{tran}(\mathsf{t}, \mathsf{s}^{\mathsf{ins}})$ and $\mathsf{tran}(\mathsf{t}^{\mathsf{rep}}, \mathsf{s}^{\mathsf{rep}})$, where $\mathsf{t}^{\mathsf{rep}}$ is the mutated phrase.

3.4 Error Detection

We identify translation errors using metamorphic testing in the error detection phase. We design three metamorphic relations (MRs) to detect term translation errors. A translation error is detected if an original sentence and its corresponding mutated sentences fail to satisfy any of the MRs.

3.4.1 MR1: The term translation of the meaning insertion mutated sentence should be consistent with the term translation of the original sentence. Given an original sentence s containing the term t, and its meaning insertion mutant $\mathsf{s}^{\mathsf{ins}}$, we obtain the translation $\mathsf{tran}(\mathsf{t}, \mathsf{s})$ of t through translation and alignment steps. Additionally, we obtain the translation $\mathsf{tran}(\mathsf{t}, \mathsf{s}^{\mathsf{ins}})$ of t. We calculate the semantic similarity between $\mathsf{tran}(\mathsf{t}, \mathsf{s})$ and $\mathsf{tran}(\mathsf{t}, \mathsf{s}^{\mathsf{ins}})$ using language model and compare it to a predefined threshold. If the similarity is below the threshold, an MR1 violation is detected.

To illustrate an MR1 violation, consider the example in Table 1. In the original sentence, the term "Part of Speech" is translated as "Part of Pronunciation." However, in the meaning insertion mutated sentence, the same term is translated accurately as "Part of Speech." The term thus has different translations in the original and mutated sentences, indicating an MR1 violation.

3.4.2 MR2: The translation of the meaning insertion mutated sentence should be consistent with the original sentence except for the explanation part. Given a original sentence s, the meaning insertion mutated sentence is $\mathsf{s}^{\mathsf{ins}}$, and their translation results are $\mathsf{tran}(\mathsf{s})$ and $\mathsf{tran}(\mathsf{s}^{\mathsf{ins}})$, respectively. Ideally, $\mathsf{tran}(\mathsf{s}^{\mathsf{ins}})$ should have the same semantics as $\mathsf{tran}(\mathsf{s})$ without including the interpretive translation of the inserted term. However, due to the uncertainty of translation results, it is challenging to eliminate the interpretive part from $\mathsf{tran}(\mathsf{s}^{\mathsf{ins}})$. Therefore, we adopt a compromise approach. Since the mutation operator inserts meaning in the form of parentheses, we match and remove all content in the Chinese and English parentheses in $\mathsf{tran}(\mathsf{s})$ and $\mathsf{tran}(\mathsf{s}^{\mathsf{ins}})$ to obtain modified translations $\mathsf{rem}(\mathsf{tran}(\mathsf{s}))$ and $\mathsf{rem}(\mathsf{tran}(\mathsf{s}^{\mathsf{ins}}))$. To ensure consistency, we reference existing MT testing work [12] and use language model to compare semantic similarity between the modified translations. If it falls below the threshold, an MR2 violation is detected.

As indicated by the MR2 violation process in Table 1, we insert the meaning of the term "Part of Speech" into the original sentence to create the mutated sentence. By comparing the two translation results, excluding the content in brackets, it is found that the translation of the meaning insertion mutated sentence is semantically similar to the original sentence translation, with the only change being the substitution of the phrase "Part of Pronunciation" with "Part of Speech." Therefore, there is no MR2 violation.

3.4.3 MR3: The term translation of the term partial replace mutated sentence should be inconsistent with the term translation of the original sentence. For an original sentence s containing a multi-word term t, its term partial replacement mutated sentence set is $\mathsf{S}^{\mathsf{rep}} = \{\mathsf{s}_1^{\mathsf{rep}}, \ldots, \mathsf{s}_n^{\mathsf{rep}}\}$, and the mutated phrases it contains are $\{\mathsf{t}_1^{\mathsf{rep}}, \ldots, \mathsf{t}_n^{\mathsf{rep}}\}$. Through the translation and alignment steps, we obtain the translation of t, $\mathsf{tran}(\mathsf{t}, \mathsf{s})$, and the translations of the mutant phrases, $\mathsf{tran}(\mathsf{t}_1^{\mathsf{rep}}, \mathsf{s}_1^{\mathsf{rep}})$, ..., $\mathsf{tran}(\mathsf{t}_\mathsf{n}^{\mathsf{rep}}, \mathsf{s}_\mathsf{n}^{\mathsf{rep}})$. Due to the stable combination of multi-word terms, the translations of the mutant phrases should differ from the translation of the original multi-word term. We calculate the minimum semantic similarity:

$$\mathsf{simi}_{\mathsf{min}} = \min\{\mathsf{Simi}(\mathsf{tran}(t, \mathsf{s}), \mathsf{tran}(t_i^{\mathsf{rep}}, \mathsf{s}_i^{\mathsf{rep}})), |i \in \{1, \ldots, n\}\}$$

where $\mathsf{Simi}(\,)$ is a similarity calculation function using language model. If the minimum similarity $\mathsf{simi}_{\mathsf{min}}$ is greater than the threshold, an MR3 violation is detected. As shown by the MR3 violation detect process in Table 1, the term in the original sentence is "Part of Speech", which is mutated to "System of Speech." After translation, the phrases are rendered with different meanings: "Part of Pronunciation" and "Pronunciation System." Therefore, there is no MR3 violation.

3.4.4 Filter error. In our method, false positives mainly stem from three sources: mismatched context for inserted meanings (MR1 and MR2), alignment tool errors (MR2), and synonymous phrases after term partial replacement (MR3). To enhance accuracy within budget and time constraints, we employ GPT to filter out detected errors. We design three prompts based on [26]: (a) GPT checks if the inserted meaning aligns with the context, filtering out errors if it doesn't; (b) GPT aligns and extracts term translations from the original and mutated sentences, filtering out MR violations if the term translations are sufficiently similar; (c) GPT determines if the original term and its mutant phrase translate to the same Chinese term, filtering out MR violations if they do.

Table 2: Dataset and Original Sentences

Field	#Sentence	#Original_Sentence
Subtitles	300,000	5,866
Science	270,000	23,024
Laws	220,000	16,925
News	450,000	42,652
Thesis	300,000	29,955
Total	1,540,000	117,929

4 EXPERIMENT SETUP

In this section, we present our experimental setup, and we have made all code and data publicly available.

4.1 Research Questions

We organize our experiments by addressing the following four research questions (RQs):

RQ1: How correct is our mutated sentence generation?

We perform metamorphic testing by mutating original sentences. Due to language complexity, these may contain errors. We randomly select and manually verify a subset of mutated sentences.

RQ2: How effectively does our method TermMT find translation errors?

In this RQ, we assess our method's effectiveness in detecting term translation errors by analyzing errors found in test cases from various domain datasets across different translation systems. We also use sampling and manual inspection to verify the precision of the detected errors.

RQ3: How does the similarity threshold impact our approach?

Our method detects errors by calculating the similarity between translation results and comparing them against a threshold. This research question aims to investigate how the choice of threshold impacts our method's performance.

RQ4: How does TermMT compare with the SOTA method?

TermTM specializes in detecting term translation errors through metamorphic testing, whereas existing metamorphic methods target overall translation inconsistency errors. This RQ aims to compare our approach with SOTA methods.

RQ5: How efficient is TermMT in terms of running time?

In this RQ, we analyze TermMT's running time, breaking down the time required for each stage of the process.

4.2 Corpus and Mutant

We extract seed sentences from the English portion of UM-Corpus [40], a comprehensive English-Chinese bilingual corpus spanning multiple domains, including Education, Laws, Microblog, News, Science, Spoken, Subtitles, and Thesis. Our experiment focuses on professional fields, including Subtitles, Science, Laws, News, and Thesis, as they feature phrase-level terms in translation contexts.

To select original sentences for mutation, we specifically choose sentences containing a single multi-word term to avoid potential error offsets (with multiple terms) that could impact error reporting accuracy. Additionally, we filter out sentences longer than 50 tokens to maintain contextual clarity. The selection outcomes of

Table 3: The Unrealistic Examples of Mutated Sentences Using Two Types of Mutation Operators

Error Type	Original Sentence	Mutated Sentence
Grammatical error (Insert Mutant)	We have of the it politicizes of the **life insurance**.	We have of the it politicizes of the **life insurance** (a form of insurance on the life of a person. If the person dies).
Counterintuitive (Insert Mutant)	What if we were to say, no more **hard rock** mining?	What if we were to say, no more **hard rock** (a rock music genre) mining?
Grammatical error (Replace Mutant)	**Milk powder** and formula foods for infant and young children–Determination of whey protein	**Milk milk** and formula foods for infant and young children–Determination of whey protein
Counterintuitive (Replace Mutant)	**fishing rod** is a stick with a hook at one end and a fool at the other	A **fishing hook** is a stick with a hook at one end and a fool at the other

original sentences are presented in Table 2, where 117,929 sentences from five fields containing single phrase-level terms are selected from the dataset of 1.54 million sentences for mutation. We ultimately generate 117,929 meaning insertion mutated sentences and 89,493 term partial replacement mutated sentences.

4.3 Studied Machine Translators

Our experiments involve evaluating both commercial and open-source translation systems. For the commercial translation models, we select Google Translate [18] and Bing Microsoft Translator [30], obtaining the translation results by calling their APIs with English-to-Chinese language settings. For the local translation model, we choose mBART [38], a multi-language NMT model supporting 50 languages. We use its fine-tuned pre-trained model with default settings from Hugging Face for English-to-Chinese translation.

4.4 Experimental Settings

In Section 3.1, we construct the dictionary using the English terminology portion of IATE [23] and Wiktionary [43] as the term thesaurus. Additionally, to extract part of speech, explanations, translations, and other term-related information from Wiktionary, we employ the Python parsing tool, wikitextparser [1].

In Sections 3.2 and 3.4, for calculating the similarity between English and Chinese sentences, we utilize the BGE [46] models (bge-base-en-v1.5 and bge-base-zh-v1.5) to represent the sentences and compute cosine similarity. These models are obtained from Hugging Face and used with their original settings. For the similarity thresholds of MR1, MR2, and MR3, we set them to 0.5, 0.7, and 0.97, respectively (refer to Section 5.3 for details on how these thresholds are determined). For the alignment process in Section 3.3, we employ the state-of-the-art alignment tool AWESOME [17] to align English terms with their Chinese translations.

Moreover, manual check is crucial for ensuring the accuracy and reliability of our results for RQs 1-4. Each item is independently labeled by two native Chinese-speaking annotators from our research team. They hold master's degrees and possess strong English proficiency. Any discrepancies are resolved through discussion to reach a consensus. The manual check results are available in the replication repository referenced in Section 9.1.

5 RESULTS

5.1 RQ1: Correctness of mutated sentences

Motivation&Approach. In this RQ, we aim to detect whether our generated mutated sentences are realistic. We randomly select 500 test cases, each containing both types of mutations, as the experimental data (1000 in total). Following previous research [42], we classify the evaluation results of each sentence into the following three categories:

- Realistic: sentences that closely resemble natural language and convey meaning accurately without significant errors.
- Grammatical error: sentences containing grammatical mistakes or structural inaccuracies.
- Counterintuitive: sentences that, despite being grammatically correct, convey a meaning that is unexpected or illogical in context.

Table 4: The Result of Correctness Manual Examination

	Insert Mutant	Replace Mutant
Realistic	98.40%	95.40%
Grammatical error	0.20%	1.60%
Counterintuitive	1.40%	3.00%
Cohen's Kappa	74.64%	85.15%

During experiments, we initially task two annotators with independently labeling and calculating Cohen's kappa coefficient. Subsequently, for sentences where annotators apply inconsistent labels, we facilitate discussion between the two annotators to resolve conflicts and achieve consistent labeling.

Results. Table 3 presents cases of grammatical and counterintuitive errors in mutated sentences. "Insert Mutant" refers to sentences mutated by the meaning insertion operator, while "Replace Mutant" refers to those generated by the term partial replacement operator. In meaning insertion mutated sentences, grammatical errors include issues like punctuation or capitalization, and counterintuitive errors occur when inserted meanings don't fit the context, such as interpreting "hard rock" as "rock music" instead of "hard stone." In term partial replacement mutated sentences, grammatical errors involve incorrect structures like "Milk milk", and counterintuitive errors arise when replacements lead to nonsensical content, like "fishing hook is a stick with a hook."

Table 4 shows the results of manual verification, with Cohen's kappa values of 74.64% and 85.15%, indicating strong consistency. Both types of mutants are over 95% realistic. Meaning insertion performs better than term partial replacement, producing fewer grammatical errors (0.20% vs. 1.60%) and counterintuitive sentences (1.40% vs. 3.00%).

Table 5: The Errors Detected Across Different Translation Systems and Datasets

MR	Models	Subtitles		Science		Laws		News		Thesis		Sum	
		#Bug	#Pct	#Bug	#Pct	#Bug	#Pct	#Bug	#Pct	#Bug	#Pct	#Bug	#Pct
MR1	Google	14	0.24%	92	0.40%	147	0.87%	108	0.25%	80	0.27%	441	0.37%
	Bing	6	0.10%	74	0.32%	96	0.57%	105	0.25%	133	0.44%	413	0.35%
	mBART	83	1.41%	462	2.01%	268	1.58%	692	1.62%	1024	3.42%	2513	2.13%
MR2	Google	19	0.32%	73	0.32%	97	0.57%	26	0.06%	24	0.08%	239	0.20%
	Bing	5	0.09%	54	0.23%	11	0.06%	21	0.05%	64	0.21%	155	0.13%
	mBART	67	1.14%	370	1.61%	77	0.45%	233	0.55%	865	2.89%	1596	1.35%
MR3	Google	60	1.02%	355	1.54%	1013	5.99%	924	2.17%	831	2.77%	3169	2.69%
	Bing	55	0.94%	272	1.18%	637	3.76%	525	1.23%	355	1.19%	1837	1.56%
	mBART	73	1.24%	386	1.68%	846	5.00%	702	1.65%	560	1.87%	2557	2.17%

Answer to **RQ1**. Both types of mutations can generate a significant proportion of realistic sentences.

5.2 RQ2: Effectiveness in Error Detection

Motivation&Approach. This section addresses RQ2, evaluating our method's ability to detect terminology translation errors. We undertake two investigations: (a) applying our approach to identify term translation bugs and (b) measuring effectiveness by manually determining the ratio of true bugs detected. We randomly select 100 errors from each machine translation output (totaling 900 errors from 3 systems across 3 MRs), which are manually inspected by two annotators, resolving any conflicts through discussion.

Results. Table 5 displays errors detected across Google Translate, Bing Microsoft Translator, and mBART systems, covering five areas: Subtitles, Science, Laws, News, and Thesis. The "#Bug" column tallies detected errors, while "#Pct" shows error proportions relative to mutant sentences in each dataset.

From Table 5, we can see that (a) Our method detects a significant number of translation errors across three machine translation systems, identifying 3,765 errors on Google Translate, 2,351 on Bing Microsoft Translator, and 6,011 on mBART. (b) Commercial-grade translation software (Google Translate and Bing Microsoft Translator) exhibits fewer translation errors compared to the translation model mBART. However, on MR3, mBART shows fewer errors than Google Translate but more errors than Bing Microsoft Translator. (c) Bing Microsoft Translator demonstrates the fewest translation errors across the three types of MR. However, in the specialized domain of thesis translation, Bing Microsoft Translator's error rate on MR1 and MR2 exceeds that of Google Translate.

Table 6 shows manual inspection results, illustrating the precision rates of the three MRs. The average precision for MR1 and MR2 error detection across the three translation systems exceeds 85%, while MR3 has a lower precision rate of 68.67%. This lower precision is due to MR3 replacing specific terminology phrases with single words, which can alter their original meaning yet still be deduced from context and thus translated the same. Additionally, the precision decrease in MR3 is influenced by unfiltered synonymous substitutions. Overall, TermMT detects translation errors with precisions of 82.33%, 83.00%, and 86.33% on Google Translate, Bing Microsoft Translator, and mBART, demonstrating that TermMT can detect errors with high accuracy.

Table 6: The Precision Values (%) of Manual Check

	MR1	MR2	MR3	Total
Bing	84.00	96.00	67.00	82.33
Google	83.00	95.00	71.00	83.00
mBART	93.00	98.00	68.00	86.33
Average	86.67	96.33	68.67	83.89
Cohen Kappa	72.28	68.37	97.68	83.37

Answer to **RQ2**. Our method identifies a substantial volume of translation errors on all three MT systems with high levels of precision.

5.3 RQ3: Impact of Similarity Threshold

Motivation&Approach. TermMT's three types of MR categorize errors based on the similarity between the translation results of the original and mutated sentences relative to a specified threshold. Consequently, the threshold setting significantly affects the quantity and precision of identified errors.

To analyze the impact of similarity thresholds and select appropriate values, we conducted experiments based on methods from existing works [22, 36]. We selected 100 original sentences from each of the three translation systems with error detection results from three MRs, totaling 900 results (3 NMT systems×3 MRs×100). Each data point was annotated by two annotators, with conflicts resolved collaboratively. We then applied a range of thresholds (0 to 1, step size 0.01) to detect errors using MRs.

Results. Figure 4 illustrates the impact of threshold settings on error quantity and accuracy across the three types of MR. The Cohen's kappa coefficients for these annotations are 80.94%, 88.62%, and 76.31%, respectively.

From Figure 4, we observe that for MR1 and MR2, increasing the threshold raises the number of detected errors but lowers precision. In contrast, for MR3, increasing the threshold decreases the number of detected errors while improving precision. TermMT, based on metamorphic testing and using an English monolingual corpus, effectively conducts high-precision error detection on large datasets. We set the similarity thresholds for MR1, MR2, and MR3 to 0.5, 0.7, and 0.97, respectively, to achieve the highest precision values. This setting is used in the error detection experiment in Section 5.2. Additionally, higher thresholds for MR1 and MR2 and

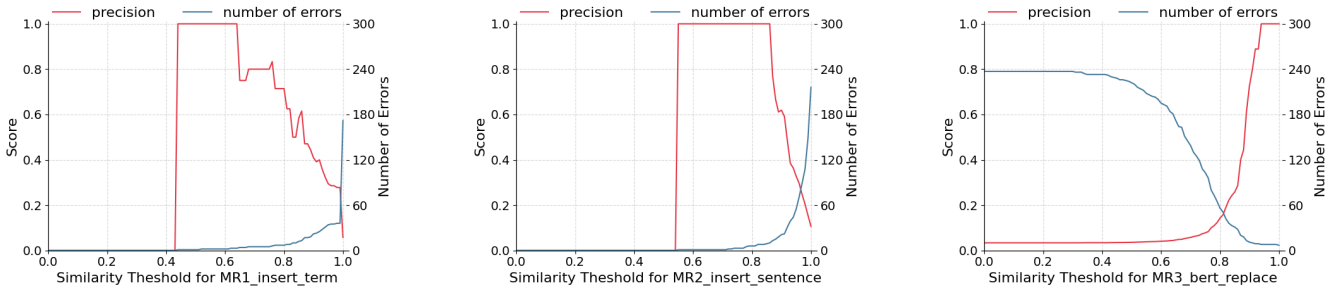


Figure 4: The Impact of Similarity Threshold when Testing Google Translate

a lower threshold for MR3 help identify a broader range of translation errors, especially useful when mutant clause sources are limited or during intensive testing phases before release [22]. This feature enhances the efficiency and practicality of our method.

> Answer to **RQ3**. The similarity threshold choice significantly impacts the number and accuracy of detected errors and can be adjusted based on specific requirements.

5.4 RQ4: Comparison with the SOTA method

Motivation&Approach. In this RQ, we compare our method with existing translation error detection methods. Due to our focus on single-phrase-level term sentences and localized mutations, direct comparisons are challenging. Therefore, we use the state-of-the-art method CAT [37] for our analysis due to its generality and ability to perform mutation and error detection on our dataset. Additionally, part of our second mutation operator aligns with CAT's word replacement approach.

CAT reports a high number of errors but with reduced accuracy due to its high threshold. Our method, however, prioritizes accuracy in threshold selection. To ensure a fair comparison, we adjust the threshold used by CAT through a threshold search experiment, similar to the methodology outlined in Section 5.3. Specifically, we randomly select 500 original sentences with error detection outcomes from CAT, verify their accuracy manually, and resolve conflicts through annotator discussion. Subsequently, we employ the dataset from Section 3.4.4 as the original set to identify translation errors in mBART using CAT. Finally, we compare the error detection performance of CAT and our method, calculating overlaps based on errors detected from the same original sentences.

Results. The results of this parameter analysis are depicted in Figure 5, with a Cohen's kappa value of 88.52% for manually labeled annotations. Based on these outcomes, we identify a threshold value of 0.6 for CAT that achieves peak accuracy while comprehensively detecting errors.

Figure 6 depicts the comparative experiment results, showing that: (a) CAT detects a similar number of errors as TermMT, with CAT detecting 8,413 and our method detecting 6,011 errors. (b) TermMT successfully identifies 1,060, 160, 1,338, 916, 1,725, and 5,168 unique errors across five datasets and in total, accounting for substantial percentages (ranging from 80.00% to 94.22%) of the errors detected by TermMT. The result underscores the effectiveness of TermMT as a valuable complement to existing testing techniques.

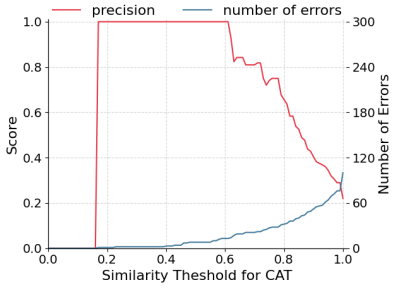


Figure 5: Impact of Threshold when Testing mBART via CAT

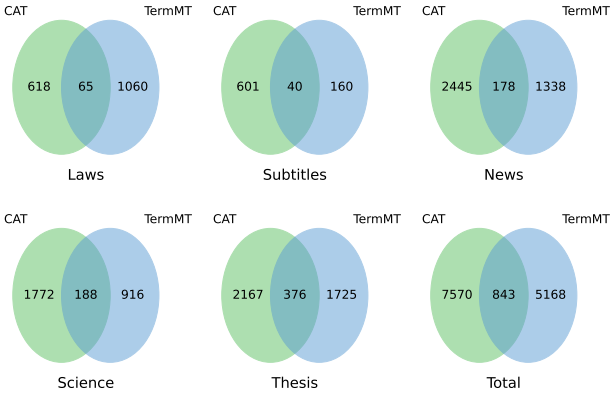


Figure 6: The Error Overlaps between TermMT and CAT

> Answer to **RQ4**. TermMT detects a comparable number of errors to CAT and identifies many unique errors across various datasets, highlighting its effectiveness as a complementary approach to existing error detection methods.

5.5 RQ5: TermMT's Efficiency in Running Time

Motivation&Approach. We evaluate the efficiency of our experiments across four key aspects: mutation generation, translation, alignment, and error detection. Experiments are conducted on a server with a 3.8 GHz CPU with 24 cores and an NVIDIA GeForce RTX 3090 GPU.

Results. The mutant generation process averages 0.21 seconds per original sentence. During translation, TermMT requires an average of 0.45, 0.52, and 2.04 seconds per original sentence on Google

Translate, Bing Microsoft Translator, and mBART, respectively. Alignment and error detection steps are minimally correlated with translation, each taking an average of 0.01 seconds and 0.04 seconds, respectively, across all translation systems. The total average time across these steps is 3.37 seconds, primarily attributed to the translation phase. Additionally, the GPT filtering process is presented separately, focusing on error processing rather than original sentences, with correct interpretation filtering averaging 0.81 seconds per target error, term translation extraction requiring 2.07 seconds on average, and synonymous replacement filtering consuming 1.06 seconds on average.

> Answer to **RQ5**. The total running time of TermMT averages 3.37 seconds per original sentence, with the translation phase being the most time-intensive.

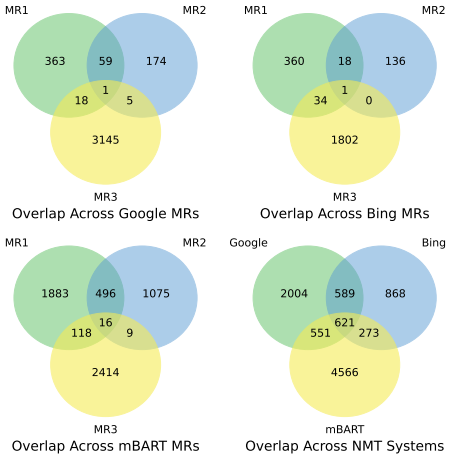


Figure 7: The Overlaps between MRs and Studied Models

6 DISCUSSION

6.1 Overlaps between MRs and Studied Models

Our method employs three MRs to analyze three distinct translation systems. In this section, we investigate the overlaps of errors detected by these MRs within each translation system and across different systems.

The experimental findings are presented in Figure 7, where the first three Venn diagrams depict the overlap of errors detected by MR1, MR2, and MR3 within the Google Translate, Bing Microsoft Translator, and mBART translation systems, respectively. In each subfigure, green, blue, and yellow ovals represent errors identified by MR1, MR2, and MR3, respectively. The final Venn diagram illustrates the overlap of errors among translation systems, with green, blue, and yellow ovals representing errors detected by TermMT on Google Translate, Bing Microsoft Translator, and mBART.

Figure 7 shows that (a) the overlap of errors among the three MRs within each translation system is minimal, suggesting that errors detected by each MR are relatively distinct; (b) the three translation systems have a considerable number of errors that overlap with other systems, and 621 errors are shared by the three translation systems, which shows that different translation systems have certain commonalities in terminology translation errors.

6.2 Contributions of GPT to the Accuracy

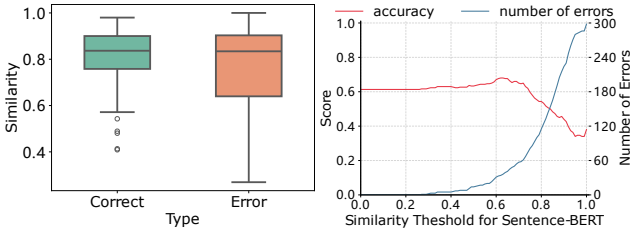
To evaluate GPT's contribution to TermMT's accuracy, we conduct an ablation experiment comparing the precision and number of detected errors with and **with**o**o**ut (denoted as TermMT_woGPT) GPT. By removing the GPT error filtering step (Section 3.4.4), we test TermMT on translation systems and the original dataset. To verify the precision of TermMT without GPT, we randomly select 100 errors from Google Translate output under each MR (totaling 300 errors across three MRs), manually inspected by two annotators.

The experimental results are as follows, mainly focusing on TermMT_woGPT: (a) TermMT_woGPT detects 5,467 errors on Google Translate (2127 by MR1, 330 by MR2, and 3191 by MR3), 3,879 errors on Bing Microsoft Translator (1,956 by MR1, 200 by MR2, and 1,857 by MR3), and 9,211 errors on mBART (5,491 by MR1, 2,198 by MR2, and 2,659 by MR3). Compared to the results in Table 5 of Section 5.2, TermMT_woGPT increases the number of detected errors. (b) For 300 sampled errors from Google Translate, TermMT_woGPT achieves a precision of 65.67% (43% on MR1, 92% on MR2, and 62% on MR3) with a Cohen's kappa value of 71.69%. This precision is slightly lower than the 83% precision of TermMT on the same dataset, as reported in Section 5.2. These results suggest that while GPT filtering enhances accuracy, TermMT remains reasonably accurate without it, indicating that GPT is not the primary contributor to TermMT's accuracy.

6.3 Performance with Different Language Model Settings for Similarity Calculation

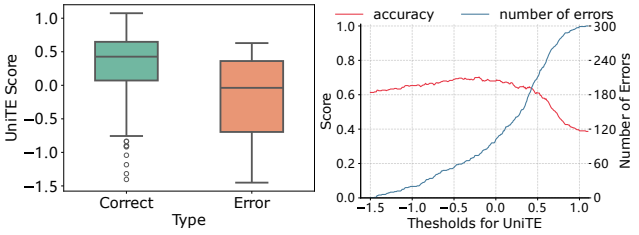
In this section, we assess TermMT's performance with language models on calculating similarity (Sections 3.2 and 3.4), focusing on two experiments: ① To evaluate TermMT's dependence on the language model, we compare it with a method that directly calculates the similarity between the original sentence and translation results, termed Language Model Detect (LMD), using the multilingual version of Sentence-BERT [34]. Since the comparison requires manual annotation, we conduct experiments on the Google Translate dataset used in Section 5.2 (i.e., 300 original-mutated sentence pairs randomly selected for 3 MRs). Since LMD does not involve mutated sentences, we only consider the original sentence, its translation, and the reference translation when re-annotating the data to determine translation errors. ② To assess the impact of using a different language model, we replace the BGE with Sentence-BERT in TermMT, resulting in TermMT_SBERT, and detect errors in Google Translate outputs. Considering Sentence-BERT's smaller model size, a potential drop in performance is anticipated. We then sample 100 manual annotations from these errors to calculate precision.

Results of the experiment ①. Figure 8 (a) shows LMD's similarity calculations on the artificial dataset, with a Cohen's Kappa of 76.84%. The close medians for correct and incorrect data indicate LMD's low accuracy in error identification. Figure 8 (b) tracks LMD's accuracy and error detection as the similarity threshold varies. Like the threshold selection strategy in Section 5.3, we set the similarity threshold at 0.63, achieving the highest accuracy 68.00%



(a) Similarity Box Plot (b) Threshold Impact on LMD

Figure 8: Evaluation of LMD Performance



(a) Score Box Plot (b) Threshold Impact on UniTE

Figure 9: Evaluation of UniTE Performance

for LMD. Consequently, LMD detects 36 errors, only 12% of the 300 errors detected by TermMT, with the remaining 88% being unique errors identified by TermMT.

Results of the experiment ②. TermMT_SBERT detects 5417 errors (287 by MR1, 234 by MR2, and 4967 by MR3), with 49% precision and a Cohen's Kappa of 65.50%, resulting in 2,654 true positives (calculated as number of detected errors×precision). In comparison, as mentioned in Section 5.2, TermMT detects 3,765 errors on the Google Translate dataset with 83% precision, yielding approximately 3,124 true positives. While Sentence-BERT increases the number of detected errors, it slightly reduces accuracy. However, the difference in the number of true positives is only about 470, demonstrating that changing the similarity calculation model has minimal impact on TermMT's ability to detect real errors.

6.4 Comparison with Reference-based Method

TermMT is designed for large-scale testing of MT systems' terminology translation without relying on reference translations, which addresses the "no test oracle" problem. This makes direct comparison with reference-based methods difficult. To mitigate this, we follow the comparison approach with LMD in Section 6.3 and conduct error detection on the same 300 re-annotated Google Translate items using the SOTA reference-based method, UniTE [41].

The box plot results in Figure 9 (a) show that while UniTE can somewhat differentiate correct from incorrect translations, the overlap in distributions and outliers among correct samples indicate its limitations in accuracy. Figure 9 (b) illustrates UniTE's error detection performance at a threshold of -0.2, achieving its highest accuracy of 73.33%. UniTE detects 75 errors, covering only 25% of the 300 errors found by TermMT, demonstrating that TermMT effectively complements UniTE in identifying errors.

6.5 How TermMT works on LLMs?

Due to the advancements in large language model technology, various commercial large language models have become widely adopted translation tools, making it pertinent to assess their efficacy in term translation. To investigate this, we randomly select 3,000 data points from the original sentence dataset and applied TermMT on one of the most popular large models, GPT-3.5 [32], for translation error detection. We utilize a straightforward translation prompt and obtained translation results by calling the OpenAI "gpt-3.5-turbo-0125" API.

The experiment results are as follows: (a) Our method detects a total of 90 errors (9 by MR1, 2 by MR2, and 79 by MR3) with a detection rate of 3%. Manual review confirms 70 of these errors (7 by MR1, 2 by MR2, and 61 by MR3) as real errors, with a Cohen's kappa value of 86.12%. The actual detection rate is 2.3%, demonstrating TermMT's effectiveness in detecting term translation errors on GPT-3.5. (b) Our method achieves an accuracy of 77.78% in detecting errors specific to GPT-3.5, underscoring its high precision and effectiveness in this context.

6.6 Translation Errors Reported by TermMT

In this section, we examine error types detected by TermMT in Google Translate and provide illustrative examples categorized by the corresponding MR. Table 7 presents the results with examples:

(a) Errors detected by MR1. In MR1, we compare term translations before and after inserting explanations to detect errors. In the original sentence, "herbal medicine" is translated simply as "medicine," while in the mutated sentence, it's accurately translated as "herbal medicine," indicating a term translation error.

(b) Errors detected by MR2. In MR2, we compare the overall sentence similarity before and after inserting explanations to detect translation errors. From Table 7, the original sentence offers a general statement, whereas the mutated sentence accurately conveys detailed GDP growth information, revealing a translation error.

(c) Errors detected by MR3. In MR3, we detect translation errors by partially replacing a word in a phrase. For instance, replacing "ethyl acetate" with "diethyl acetate" results in both being translated as "ethyl acetate," indicating a term translation error.

6.7 The Insights of Our Study

Our method, TermMT, achieves translation system error detection without reference translations by designing mutation operators and metamorphic relations around terminology features. Experiments demonstrate that TermMT can detect a significant number of errors across three translation systems using three MRs. This approach may offer insights for the following future studies:

(a) *Terminology-Specific Testing.* By focusing on the unique challenges posed by terminology translation, our method highlights the importance of developing specialized testing techniques tailored to different aspects of language translation.

(b) *Importance of Terminology Translation Testing.* Accurate terminology translation is crucial in professional and technical fields, as errors can lead to significant misunderstandings and miscommunications. Such importance underscores the need for specialized testing methods to ensure translation precision and clarity.

Table 7: Examples of Translation Errors Detected by TermMT in Google Translate

MR	English (Source Language)	Chinese (Target Language)	Target Meaning
MR1	Original Sentence. Therapeutic effect on T-lymphocyte sub-population in cor-pulmonal with **herbal medicine** Astragalus membranaceus (xisch) Bge and ligusticum chanxiong Hort	黄芪、川芎**药**对肺心病 T 淋巴细胞亚群的治疗作用	Therapeutic effect on T-lymphocyte sub-population in cor-pulmonal with **medicine** Astragalus membranaceus (xisch) Bge and ligusticum chanxiong Hort
	Mutated Sentence. Therapeutic effect on T-lymphocyte sub-population in cor-pulmonal with **herbal medicine** (herbalism) Astragalus membranaceus (xisch) Bge and ligusticum chanxiong Hort	**中药**黄芪、川芎对肺心病 T 淋巴细胞亚群的治疗作用	Therapeutic effect on T-lymphocyte sub-population in cor-pulmonal with **herbal medicine** Astragalus membranaceus (xisch) Bge and ligusticum chanxiong Hort
MR2	Original Sentence. **Cape Verde**, over the same four years, recorded annual average GDP growth of 2.3 percent and in 2004 is slightly below the average of the 50 least developed countries, with growth of 5.5 percent.	**佛得角**财政部长姆努钦表示，佛得角的经济增长潜力巨大，预计今年将达到 100 万亿美元。	Cape Verde's Finance Minister Mnuchin said Cape Verde's economic growth potential is huge and is expected to reach US$100 trillion this year.
	Mutated Sentence. **Cape Verde** (archipelago in West Africa), over the same four years, recorded annual average GDP growth of 2.3 percent and in 2004 is slightly below the average of the 50 least developed countries, with growth of 5.5 percent.	**佛得角**（西非群岛）在同一四年中，国内生产总值年均增长率为 2.3%，2004 年略低于 50 个最不发达国家 5.5%的平均增长率。	**Cape Verde** (archipelago in West Africa), over the same four years, recorded annual average GDP growth of 2.3 percent and in 2004 is slightly below the average of the 50 least developed countries, with growth of 5.5 percent.
MR3	Original Sentence. Having higher yield (0.40%~0.60%) and purity(97%~98%),the use of **ethyl acetate**/water system is obviously advantageous.	采用**乙酸乙酯**/水体系具有明显的优势，收率 (0.40%~0.60%) 和纯度 (97%~98%) 较高。	Having higher yield (0.40%~0.60%) and purity(97%~98%),the use of **ethyl acetate**/water system is obviously advantageous.
	Mutated Sentence. Having higher yield (0.40%~0.60%) and purity(97%~98%),the use of **diethyl acetate**/water system is obviously advantageous.	采用**乙酸乙酯**/水体系具有明显的优势，收率 (0.40%~0.60%) 及纯度 (97%~98%) 较高。	Having higher yield (0.40%~0.60%) and purity(97%~98%),the use of **ethyl acetate**/water system is obviously advantageous.

7 RELATED WORK

This section presents two points of related studies, Machine Translation Testing and Metamorphic Testing.

Machine Translation Testing. Traditional translation tests rely on bilingual corpora and various machine indicators like BLEU [33], METEOR [6], and ROUGE [27] to evaluate translation performance. However, creating high-quality bilingual corpora is challenging [20]. To overcome this problem, recent methods circumvent the need for reference translations. Aiken et al. [2] introduced round-trip translation testing (RTT). He et al. [22] proposed structural invariance testing (SIT), assuming translations should maintain sentence structure after word replacement. Sun et al. [36] developed TransRepair, which uses mutation and metamorphic testing, later refining this with the CAT method using BERT for contextual consistency [37]. Zhang et al. [48] employed Syntactic Tree Pruning for term translation metamorphic testing, using a bag-of-words model to detect translation errors based on sentence similarity.

Metamorphic Testing. Metamorphic Testing (MT) is a black-box testing method that addresses the oracle problem by using metamorphic relationships (MRs) to describe the expected properties between inputs and outputs [15]. This method can be applied to unlabeled datasets and is widely used in AI testing. For instance, Asyrofi et al.[5] proposed BiasFinder, which detects bias in sentiment analysis by generating mutated sentences. Zhang et al. [47] employed metamorphic testing to evaluate autonomous driving systems (Deeproad) with diverse weather conditions and validated input images. Chen et al. [14] applied metamorphic testing to question-answering systems (qaAskeR) by designing three metamorphic relations to examine QA software's responses to multiple recursive questions related to the same knowledge.

8 THREATS TO VALIDITY

Threats to External Validity. First, the selection of translation systems presents a potential threat. We primarily test Google Translate, Bing Microsoft Translator, and mBART, with a brief discussion on GPT-3.5. A broader evaluation of more translation systems and large models is needed. Although we used neural-based translation methods as the test subjects, our method TermMT, as a black-box testing approach, can be easily adapted to other translation systems, including rule-based systems. Second, our focus was on English-to-Chinese translation. Given our method only requires a monolingual corpus, it is feasible to test various language pairs using TermMT. Future work will extend this exploration.

Threats to Internal Validity. Internal threats mainly stem from the manual review process. We follow manual review settings from existing studies [36, 37]. To minimize participant bias, we provide detailed introduction documents and keep the source of issues revealed by TermMT and the baseline invisible to participants. The high Cohen kappa value indicates minimal bias in manual evaluation. Additionally, we have made our manual results publicly available on GitHub to enhance transparency and replicability.

9 CONCLUSION

In this paper, we focus on evaluating the terminology translation capabilities of translation systems and propose a novel metamorphic testing method, TermMT, for large-scale testing in unlabeled scenarios. TermMT detects translation errors through mutation operators and metamorphic relations based on the characteristics of terms. Our experimental results demonstrate that TermMT can accurately detect a significant number of translation errors. In the future, we plan to develop additional metamorphic relations and expand our testing to include more translation systems.

9.1 Replication Package

We provide data and source code used to conduct this study at https://github.com/mt4ai/TermMT.

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